CURRICULUM VITAE



CHRISTOPHER A. MILLER, PH.D.

DATE: July 1, 2026

CITIZENSHIP: United States of America

ADDRESS AND TELEPHONE NUMBERS:
Campus Box 8501
660 South Euclid Ave,
St. Louis, Missouri 63110

PRESENT POSITION: Associate Professor, Division of Oncology,
Washington University School of Medicine

EDUCATION:

May 2005	B.S. Biology and Computer Science Truman State University, Kirksville, MO
May 2011	Ph.D. Structural and Computational Biology and Molecular Biophysics Program Baylor College of Medicine, Houston, TX Thesis topic: Computational Cancer Genomics

ACADEMIC POSITIONS:

2011-2013	Staff Scientist - Medical Genomics, The Genome Institute Washington University in St Louis
2013-2017	Cancer Analysis Group Leader, The Genome Institute Washington University in St Louis
2015-2016	Instructor in Medicine, Division of Genomics and Bioinformatics Washington University School of Medicine
2016-2022	Assistant Professor in Medicine, Division of Oncology Washington University School of Medicine
2022-Present	Associate Professor in Medicine, Division of Oncology Washington University School of Medicine

MEDICAL LICENSURE AND BOARD CERTIFICATION:

HONORS AND AWARDS:

EDITORIAL/REVIEW RESPONSIBILITIES:

- NIH Study Section reviewer - MCTB - Mechanisms of Cancer Therapeutics B

- Reviewer for *Bioinformatics*, *Cancer Cell*, *Genome Biology*, *PLoS Computational Biology*, *Nucleic Acids Research, Nature Communications, Genome Research, Blood Cancer Discovery, Scientific Data, BMC Evolutionary Biology, BMC Bioinformatics, Journal of Open Source Science, et al.*
- Grant reviewer for Dutch Cancer Society (KWF Kankerbestrijding)
- Grant reviewer for Siteman Cancer Center Investment Fund

PROFESSIONAL SOCIETIES:

American Association for Cancer Research
International Society for Computational Biology
Cancer Genomics Consortium

Oral Presentations (conference - title):

2012	TCGA Symposium - Assessing Tumor Heterogeneity and Tracking Clonal Evolution Using Whole Genome or Exome Sequencing
2013	TCGA Steering Committee Meeting - Automatic inference of theraputically relevant subclones in heterogeneous tumors
2013	Precision Medicine: Personal Genomes and Pharmacogenomics – SciClone: Inferring clonal architecture and tracking the spatial and temporal patterns of tumor evolution and therapy resistance
2014	TCGA Symposium - Ultra-deep whole-genome sequencing reveals clinically relevant low-frequency subclones in an acute myeloid leukemia
2015	Genome Informatics - Assessing tumor heterogeneity and tracking clonal clearance in response to therapy
2016	Festival of Genomics - Genomic insights into tumor evolution and therapy response
2016	Genome Informatics - Characterizing genomic responses to cancer immunotherapy
2019	Genomics in Medicine - Tumor Heterogeneity and Evolution
2020	Genomics Education Partnership (GEP) Workshop - Sequencing, Variant Calling, and Cancer Genomics
2021	Institute of Medical Informatics in Münster, Germany - Genomic and Transcriptional evolution of AML at single-cell resolution
2021	Truman State University Biology Seminar Series - Computational cancer genomics - using "Big Data" to understand how tumors evolve
2023	Truman State University Computer Science Colloquium - Computer Science in biology: using bioinformatics to understand and treat cancer
2025	Southern Illinois University Biology Colloquium - Cancer Genomics and Bioinformatics: Using computational methods to better understand and treat AML

2026	Webster University: Using big data and multi-omics to identify therapeutic opportunities in cancer
2026	Truman State University: Cancer Genomics and Bioinformatics: Using computational methods to better understand and treat AML

RESEARCH SUPPORT:

1 R50 CA211782 (PI)
NIH-NCI 09/01/2017-08/31/2027
Title: Comprehensive informatic analyses of AML genomes and epigenomes

5 R35 CA197561 (Analyst - PI: Ley)
NIH-NCI 08/15/2015 - 06/30/2029
Title: Molecular Pathogenesis of Acute Myeloid Leukemia

TEACHING TITLE AND RESPONSIBILITES:

2005	Teaching Assistant - Advanced Topics: Quantitative Biology (BCM)
2007-2008	Teaching Assistant - Computer Aided Discovery Methods (BCM)
2008	Teaching Assistant - Computational Mathematics for Biomedical Scientists (BCM)
2015-2017	Lecturer - Investigating Eukaryotic Genomes (WUSTL)
2015-2019	Lecturer - Molecular Basis of Heredity (WUSTL)
2017-2021	Lecturer - Computational Statistical Genetics (WUSTL)
2018	Lecturer - Research Explorations in Genomics (WUSTL)
2019-2021	Lecturer - Human Genetics (WUSTL)
2019	Lecturer - High-Throughput Biology: From Sequence to Networks (CSHL/CBW Course)
2017-	Organizer - Applied Bioinformatics for Genomics I and II (formerly WUSTL Bioinformatics Workshop) - full academic year course, developed curriculum, recruited faculty - designed lectures on sequencing, informatics, somatic variant calling, bisulfite sequencing and analysis, bash, HPC/LSF/Docker, etc
2021-	Organizer - Computational Cancer Genomics Summer Program for undergraduate/masters students.
2021-	Organizer - Bioinformatics Bootcamp - teaching biology and bioinformatics to undergraduate/masters summer students in a "crash course" format.
2021-	Co-Organizer - Advanced Sequencing Technologies & Bioinformatics Analysis (Cold Spring Harbor Laboratory course)

2023	Truman State University. Invited guest lecture on Bioinformatics
2024	University of Missouri - Rolla. Invited guest lecture on Cancer Genomics and Bioinformatics
2025-	Lecturer - Genomics in Medicine Seminar Series (WUSTL)

TRAINEE RECORD:

Ph.D Thesis Committee member

Katie Campbell	2016-2018
Cody Ramirez	2017-2021
Miller Richters	2019-2023
My Hoang	2024-
Mariam Khanfar	2024-

Mentor to undergraduate research students

Yang Li	2013	Software Engineer at Emory University
Callie Federer	2014	PhD, now scientist at RadiaSoft
Jonas Neichen	2017	Medical Student at Rush Medical College
Megan Neveau	2017	PhD, now DVM student at Iowa State University
Zhaohe Zhao	2020	Instructor, Chengdu University, China
Andrew Li	2021	Software Engineer, Capital One
Payton Redemer	2022	Software Engineer, Garmin
Nguyen Le	2023	Graduate Student, University of Melbourne
Van Grabner	2024	Student at University of Missouri
Brady Maher	2025	Student at UMSL
Tianyi Fu	2025	Staff Scientist at WUSTL

Training and Supervision of WUSTL computational staff, including

Dr. Gue Su Chang	2013-	Cancer Analysis Group, TGI
Dr. Tiandao Li	2013-	Cancer Analysis Group, TGI
Dr. Allegra Petti	2013-16	Cancer Analysis Group, TGI
Dr. Yevgeniy Gindin	2014-15	Cancer Analysis Group, TGI
Dr. Charles Lu	2013-15	Cancer Analysis Group, TGI
Sai Mukund Ramakrishnan	2019-26	AML PPG Core C/Ley Lab Staff

Additional formal and informal bioinformatics training of dozens of internal and external collaborators

BIBLIOGRAPHY:

(* indicates co-first authorship)

1. Cancer Genome Atlas Network. Comprehensive genomic characterization defines human glioblastoma genes and core pathways. Nature. 2008 Oct 23;455(7216):1061-8. doi: 10.1038/nature07385. Epub 2008 Sep 4. PMID: 18772890 PMCID: PMC2671642
2. Hampton OA, Den Hollander P, **Miller CA**, Delgado DA, Li J, Coarfa C, Harris RA, Richards S, Scherer SE, Muzny DM, Gibbs RA, Lee AV, Milosavljevic A. A sequence-level map of chromosomal breakpoints in the MCF-7 breast cancer cell line yields insights into the evolution of a cancer genome. Genome Research. 2009 Feb;19(2):167-77. doi: 10.1101/gr.080259.108. Epub 2008 Dec 3. PMID: 19056696 PMCID: PMC2652200
3. Coarfa C, Yu F, **Miller CA**, Chen Z, Harris RA, Milosavljevic A. Pash 3.0: A versatile software package for read mapping and integrative analysis of genomic and epigenomic variation using massively parallel DNA sequencing. BMC Bioinformatics. 2010 Nov 23;11:572. doi: 10.1186/1471-2105-11-572. PMID: 21092284 PMCID: PMC3001746
4. **Miller CA**, Hampton O, Coarfa C, Milosavljevic A. ReadDepth: a parallel R package for detecting copy number alterations from short sequencing reads. PLoS One. 2011 Jan 31;6(1):e16327. doi: 10.1371/journal.pone.0016327. PMID: 21305028 PMCID: PMC3031566
5. **Miller CA**, Settle SH, Sulman EP, Aldape KD, Milosavljevic A. Discovering functional modules by identifying recurrent and mutually exclusive mutational patterns in tumors. BMC Med Genomics. 2011 Apr 14;4:34. doi: 10.1186/1755-8794-4-34. PMID: 21489305 PMCID: PMC3102606
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7. Parnell LD, Lindenbaum P, Shameer K, Dall'Olio GM, Swan DC, Jensen LJ, Cockell SJ, Pedersen BS, Mangan ME, **Miller CA**, Albert I. BioStar: an online question & answer resource for the bioinformatics community. PLoS Computational Biology. 2011 Oct;7(10):e1002216. doi: 10.1371/journal.pcbi.1002216. Epub 2011 Oct 27. PMID: 22046109 PMCID: PMC3203049
8. Ding L, Ley TJ, Larson DE, **Miller CA**, Koboldt DC, Welch JS, Ritchey JK, Young MA, Lamprecht T, McLellan MD, McMichael JF, Wallis JW, Lu C, Shen D, Harris CC, Dooling DJ, Fulton RS, Fulton LL, Chen K, Schmidt H, Kalicki-Veizer J, Magrini VJ, Cook L, McGrath SD, Vickery TL, Wendl MC, Heath S, Watson MA, Link DC, Tomasson MH, Shannon WD, Payton JE, Kulkarni S, Westervelt P, Walter MJ, Graubert TA,

Mardis ER, Wilson RK, DiPersio JF. Clonal evolution in relapsed acute myeloid leukaemia revealed by whole-genome sequencing. Nature. 2012 Jan 11;481(7382):506-10. doi: 10.1038/nature10738. PMID: 22237025 PMCID: PMC3267864

9. Ellis MJ, Ding L, Shen D, Luo J, Suman VJ, Wallis JW, Van Tine BA, Hoog J, Goiffon RJ, Goldstein TC, Ng S, Lin L, Crowder R, Snider J, Ballman K, Weber J, Chen K, Koboldt DC, Kandoth C, Schierding WS, McMichael JF, **Miller CA**, Lu C, Harris CC, McLellan MD, Wendl MC, DeSchryver K, Allred DC, Esserman L, Unzeitig G, Margenthaler J, Babiera GV, Marcom PK, Guenther JM, Leitch M, Hunt K, Olson J, Tao Y, Maher CA, Fulton LL, Fulton RS, Harrison M, Oberkfell B, Du F, Demeter R, Vickery TL, Elhammali A, Piwnica-Worms H, McDonald S, Watson M, Dooling DJ, Ota D, Chang LW, Bose R, Ley TJ, Piwnica-Worms D, Stuart JM, Wilson RK, Mardis ER. Whole-genome analysis informs breast cancer response to aromatase inhibition. Nature. 2012 Jun 10;486(7403):353-60. doi: 10.1038/nature11143. PMID: 22722193; PMCID: PMC3383766.
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